ARYAN SRIVASTAVA

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PROFILE

Second-year Computer Science student focused on CUDA-adjacent GPU systems, GPU dataframe internals, LLM inference benchmarking, backend infrastructure, and production-grade open-source development.
Contributes to large C++/Python and Rust codebases, writes tests, fixes behavioral inconsistencies, and works across backend, AI, and HPC systems.

EDUCATION

Manipal Institute of Technology, Bengaluru
B.Tech in Computer Science and Engineering | Expected January 2028

OPEN SOURCE CONTRIBUTIONS

NVIDIA RAPIDS - cuDF
- PR #20747: Boolean Casting Consistency Fix, merged.
- Fixed behavioral inconsistency between cuDF and Pandas; modified Python/C++ interop logic and added regression tests.
- PR #20862: Hybrid Scan API, merged.
- Added a hybrid scan API in libcudf C++ for all-true row masks with memory-safe null handling and unit tests.

Polars - High-Performance DataFrame Library (Rust)
- PR #27669: FixedRingBuffer Allocation Provenance, merged.
- Fixed a Rust allocation-provenance soundness issue in polars-utils by preserving the original Vec allocation.

PROJECTS

MAHE Mobility Challenge 2026 - 1st Place, AI Track | Backend Architect
- Built a connected-vehicle alert pipeline with local Phi-3 classification, Node.js/WebSocket triage, a heatmap, and recovery summaries.

LLM Inference Benchmarking | Sarvam 30B FP8
- Repo: github.com/aryansri05/indicservebench
- Benchmarked Sarvam 30B FP8 across NVIDIA H100 SXM, NVIDIA T4, and Apple M2 for real-world inference performance.
- Measured mean latency, P90/P95 latency, tokenization behavior, throughput trends, and hardware deployment tradeoffs.

Paper Trading Platform
- Full-stack options simulator with React, Node.js/Express, PostgreSQL, JWT auth, portfolio tracking, and real-time market data.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL, Bash
AI / ML: LangGraph, FastMCP, ChromaDB, Ollama, Llama 3, RAG, LLM agents
HPC / GPU: CUDA, cuDF, RAPIDS AI, LLM inference benchmarking, Apple Silicon unified memory, SLURM
Frameworks: React.js, Node.js, Express, Pandas, NumPy, Pytest
Tools: Git, GitHub Actions, Docker, CI/CD, PostgreSQL, GraphQL, VS Code